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BCvoteMDE: A Blockchain-based E-Voting Scheme for Multi-District Elections

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***Abstract*—The traditional electronic voting (e-voting) has the problems of dead vote, repetition and missing registration, which cannot reflect the voting result correctly, objectively and fairly. With the booming development of blockchain in recent years, blockchain technology provides a new solution in the field of e-voting. Multi-District election is an important method of election in real life, e.g. in US presidential election. Inspired by the electoral college system, we propose a blockchain-based e-voting scheme for multi-district election. First, we propose a blockchain-based voter registration method that all voters are authenticated by the blockchain system. Second, we design a two-layer blockchain architecture, where the lower layer records the votes of voters in each district and the higher layer records the votes of electors. Then we describe the voting process and evaluate the security and availability of the proposed scheme. The experimental results show that the proposed scheme can satisfy the needs of multi-district election.**

Keywords—Blockchain, Electronic Voting, Multi-District Elections.

I. Introduction

Election is a form of procedure with recognized rules which is almost conducted through voting. Given the importance of elections to society, efforts have been made to optimize the reliability and efficiency of elections. With the development of technology, voting has evolved from traditional paper-based voting to electronic voting. Traditional Paper-based voting system used paper ballots for voting and counted by hand, while the E-voting system used the electronic voting in an Internet environment. The efficiency of voting has been increasing while the cost of voting has been decreasing. However, both paper-based and electronic voting are centralized voting systems, the centralized structure suffers from single point of failure as well as can lead to trust issues, such as double voting, dead ballots, lost ballots, deliberate delays in counting, inefficiency tallying, etc. In addition, the ballots recorded in traditional voting systems can be easily changed or even deleted. A real example is the South Wales (NSW) Electoral Commission's admission that the online iVote website [1] leaked as many as 66,000 electoral votes.

Blockchain technology brings a new solution to electronic voting systems. Blockchain has received considerable attention since its introduction in 2008 [2]. Blockchain is a distributed ledger, which is decentralized, tamper-proof and traceable. Initially used in the field of cryptocurrency, blockchain has since been applied to a variety of fields such as supply chain [3], healthcare [4], IOT [5], and more. Exploiting these exclusive properties, blockchain are also suitable for application to the field of electronic voting. It is believed that applying blockchain to the field of e-voting has great prospects. Electronic voting schemes based on blockchain technology have been a hot topic of research in recent years, and some results in this area have emerged.

The indirect election under multiple election districts is rarely discussed in existing research works [7], [8], [13],[15], [17]. Most of them just realize the single stage voting with single district for a small scale election. In fact, multi-district elections are widely adopted in various elections. Countries with complex regional electorates may elect a head of state by means of an electoral college rather than a direct popular election. A contemporary political example is the use of electoral college system for presidential elections in the United States. Looking back at the 2020 U.S. presidential election, due to the drawbacks of paper-based voting many people believe that there was fraud in the election after the results were published. From the evidence of election fraud that has come to light so far, it includes a large number of fake ballots, normal ballots being lost, expired ballots being counted, the counting process evading supervision, and tampering with counting data. Blockchain technology is expected to solve these shortcomings. Therefore, it is necessary to introduce the electoral college system into the field of electronic voting.

Inspired by these challenges, we are motivated to construct a blockchain-based e-voting scheme with two-layer architecture supporting Multiple Electoral District Mechanism. In a single election using the two-stage multiple election districts mechanism, each district is relatively independent and conducts the election simultaneously. In the final election stage, it is not necessary to know the specific voting contents of each district, but only the election results of each district. Compared with the existing blockchain, the two-layer blockchain architecture can firstly meet the above-mentioned characteristics of multi-district election and improve the scalability easily. Our contributions can be summarized as follows:

- We propose a blockchain-based e-voting scheme for election scenarios for multi-district elections.
- We propose a two-layer blockchain architecture for voting to satisfy the election requirements of the electoral college system.
- We optimize the voting process and improve the efficiency of voting.

The rest of the paper is organized as follows: Section II presents related work of e-voting systems based on blockchain. Section III gives an overview of the voting system and describes the goals of the system implementation. The composition and block structure of the two-layer blockchain architecture is explained in Section IV. The voting process is illustrated in Section V and the conclusion can be seen in Section VI.

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II. Related Work

Electronic voting systems are often implemented on cryptocurrencies, and the general idea of these systems is to use cryptocurrencies as ballots in voting system. Building e-voting system based on the Bitcoin is the approach adopted by some literature [6,7]. In [7], a decentralized end-to-end voting platform based on Bitcoin was presented. Zhao proposed a Bitcoin-based electronic voting system and used zero-knowledge proof technology in the system, which protects personal privacy while voting [8]. However, the voting system built on top of Bitcoin has the limitations due to the long period of consensus building and the large amount of computing power required. Other electronic voting platforms are implemented on Ethereum [10-15]. Ethereum was proposed in 2013 by programmer Vitalik Buterin. Today Ethereum is the second-largest cryptocurrency by market capitalization after Bitcoin. As the world's largest open platform, the widespread use of Ether and provision of smart contracts logic are the reasons why many researchers have chosen it to build electronic voting systems. In particular, the extensive use of smart contracts has led researchers to recognize the ability to apply smart contracts to the field of electronic voting to eliminate human influence in the voting process. In [10], the author implemented and tested an e-voting application based on private Ethereum network using smart contracts. Nevertheless, the application only focuses on small-scale elections and polls. The author of [10] implemented a self-tallying voting system using smart contracts. McCorry and others presented a two-round protocol based on Ethereum which was used smart contracts to improve voter privacy [15]. The voting protocol is the first implementation that does not rely on any trusted authority to compute the tally. But this protocol only supports two election options and small-scale voting.

Furthermore, some researchers did not choose to build an electronic voting system on the top of existing blockchain projects, but chose to modify the current blockchain architecture to adapt to the electronic voting system. In [16], Abuidris and others proposed a hybrid consensus (PSC-Bchain) composed of Proof of Credibility and Proof of Stake combined with sharding to improve the scalability and performance of e-voting systems. The author of [17] also presented a e-voting system based on Ethereum. Unlike other solutions, their system is built on Ethereum private blockchain and uses district-based voting. Furthermore, Ayed designed a conceptual e-voting solution [18]. Ayed modified the structure of the blockchain by writing the candidates' information into the Genesis block, and the ballots for that specific candidate are linked after the candidate block thus forming multiple candidate blockchains. However, the article only presents the conceptual structure without elaborating its details. In [19], Barnes and others proposed a e-voting system using two distinctly separate blockchains. They im-proved voter anonymity when selecting their vote through the two distinct block-chains. After that, the author of [20] also implemented a e-voting with a dual-blockchain architecture recently. They utilized separate chains to maintain voter secrecy.

III. The Voting System for Multiple Electoral District Mechanism

In this section, we provide an overview of our e-voting scheme based on consortium blockchain which is shown in Fig.1. It describes the objectives of the system design and the various entities in the system. The voting scheme uses a two-layer architecture to functionally satisfy the election requirements of the multi-district electoral mechanism. We believe that formal elections and voter information should belong to the government (or election committee), which is different from other non-official voting forms. Therefore, election administrator and national election authority are introduced into the proposed voting system.

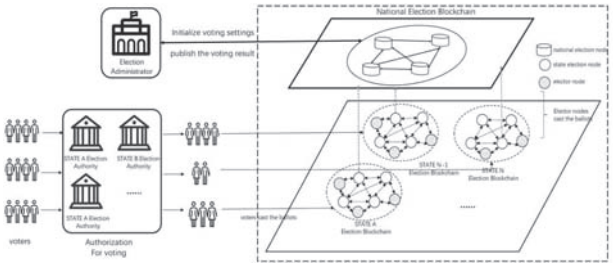

Fig. 1. The architecture of e-voting scheme based on blockchain for Multi-District Elections.

A. Design Objective

In order to build a system that can be used in large-scale e-voting, we discussed the multi-district voting. And inspired by electoral college, our blockchain voting system first meet the requirements of the electoral college, which we summarize as the winner-take-all method and multi-district voting. In addition to that, the system should meet the requirements of a large-scale electronic voting system.

B. The electoral college system

Inspired by the U.S. electoral college system, we have extracted two key mechanism from the electoral college system, we introduce the two mechanism we use in our work: multi-district elections and winner-take-all:

- **multi-district voting:** There is a set of candidates $C = \{c_0, c_1\}$ and a set of voters $V = \{v_1, \ldots, v_n\}$ divided in a set D of districts. The set of voters of district $d \in D$ is denoted with v^d. Each voter casts a vote for one of the two candidates.
- **winner-take-all:** For each $d \in D$, the result of a district's election represents the will of all voters in that district, i.e., the ballots of the district are given to the candidate who receives a relative majority of votes in that district.

C. Involved Entities

As illustrated in Fig.1, there are four types of entities involved in our voting system. Respectively, these four entities are the voters, the national election authority, the election administrator, the blockchain network for voting. The details of these entities are explained as follows

- Voters: The Voters are the group of people in the system who are eligible to cast a ballot. Voters need to generate a secret key pair and be certified by the election authority.
- National election authority (NEA): The national election authority is responsible for verifying that users of the system are eligible to vote in this election. The national election authority will distribute certificates if the users are qualified.
- Election administrator (EA): The election administrator is responsible for initiating an election

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and initializing the settings in the blockchain network. The settings include candidate list, election id, election start time and end time.

- Blockchain network for voting: The proposed blockchain network for voting is based on a two-layer framework consortium blockchain. The lower layer is the state election blockchain and the higher layer is the national election blockchain. The state election blockchain is divided into several corresponding blockchains according to electoral district, and each state election blockchain only accepts votes from the voters of that state. The state election blockchain is responsible for accepting the votes of voters in each state, while the national election blockchain is used not only for auditing votes and casting votes in the elector nodes, but also for counting votes and announcing results. Such a two-layer blockchain architecture can reduce the use of smart contracts by placing the functionality on the application system. In addition, the two-layer architecture can satisfy the winner-take-all rule and multi-district elections, so it is very suitable for the multi-district elections voting scenario.

IV. The Two-Layer Blockchain Architecture for Voting

To better satisfy the needs in the multi-district elections scenario, we propose a two-layer blockchain architecture for voting as illustrated in Fig.1. This architecture consists of a national election blockchain on the upper layer and several election blockchain on the lower layer. All blockchains in the system are consortium blockchains. In this section, we describe in detail how this two-layer blockchain architecture works and how it can be applied to elections.

A. Role of Participants

The lower layer of the two-layer blockchain architecture consists of several state election blockchains, and each state election blockchain is established by each state election committee. Each state election blockchain represents an electoral district, so each state election blockchain only accepts votes from voters in that state (or electoral district). Each state election blockchain contains two kinds of nodes, one is the state election node and the other is the elector node. The state election node participates in the consensus process and is responsible for collecting voters' ballots, verifying the compliance of the ballots, and creating blocks. The elector node represents the electors in the electoral college system. The elector node is the bridge that connects the state election blockchain to the national election blockchain. Multiple elector nodes exist in each state, and the elector nodes are only responsible for the corresponding state election blockchain. Each elector node has the right to vote once in an election. The elector node does not participate in the consensus of any layer of the blockchain; its responsibility is to verify the election results of the corresponding state after an election and cast its own elector ballot to the state election node according to the winner-take-all method.

B. The generation of Two-layer Blockchain Architecture

Before the election begins, all consensus nodes and elector nodes participating in the election need to be divided by electoral district. Consensus nodes are randomly assigned to a electoral district through a security algorithm called Verifiable Random Function (VRF). Verifiable Random Function (VRF) was proposed by Micali[21]. Only the holder of the VRF private key can calculate the hash, but anyone with the corresponding public key can verify the correctness of the hash. VRF can realize non-interactive lottery. We use the VRF to randomly allocate nodes to district. Each state has some consensus nodes, called state election nodes if assigned to the underlying election blockchain, or national election nodes if assigned to the national election blockchain. The number of elector nodes in each state election blockchain is the base number plus the additional number. The additional number is determined in advance by the population of the electoral district and is proportional to the population of the electoral district. The algorithm can be seen as follows.

Algorithm 1. Allocate consensus nodes according to election districts

Input: m: the number of states, $Start\ Timestamp$: election start timestamp, SK_i: private key of $node_i$, VRF: Verifiable Random Function, $NList$: the list of consensus node, $DList_i$: the list of $district_i$ consensus node
Output: result of allocation
1: **Begin:**
2: **for** each $node_i$ in $NList$ **do**
3: $< R_i,\ P_i >= VRF_{hash}(SK_i, StartTimestamp)$
4: $n = R_i \bmod m$
5: $DList_n$.append($node_i$);
6: broadcast $(R_i,\ P_i)$ to all elector nodes
7: **End for**
8: **End**

C. The Structure and Building of Block Data

We modify the blockchain data structure in the blockchain to apply to the voting system. The data stored in the lower-level state election blockchain is the ballots of the voters in each state. The header of the block mainly includes the block number, timestamp, block hash, previous block hash, version number, election id, candidate id, current state id, election start timestamp, and election end timestamp. The block's block body holds the voter's ballot. The actual number of ballots stored in the block body should be determined by the number of voters. If there are more voters then more ballots should be stored in each block body, while if there are fewer voters then fewer ballots should be stored in the block body. The number of ballots in each block body should be the same in an election. Table 1 gives an example of the block structure of state election blockchain.

TABLE I. Example of block structure of state election blockchain

Block Header	block number	timestamp	block hash	previous block hash	version number
	election id	candidate id	current state id	election start timestamp	election end timestamp
Block Body	voter's ballot	voter's ballot	…	…	voter's ballot

The block structure in the national election blockchain is similar to that in the state election blockchain, where the block headers have the same structure, but the block body in the national blockchain holds only one elector's ballot for the

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candidate. Table 2 gives an example of the block structure of national election blockchain.

TABLE II. EXAMPLE OF BLOCK STRUCTURE OF NATIONAL ELECTION BLOCKCHAIN

Block Header	block number	timestamp	block hash	previous block hash	version number
	election id	candidate id	current state id	election start timestamp	election end timestamp
Block Body	elector's ballot				

Since under the winner-take-all method, all of the state's electoral votes are given to the candidate who receives the relative majority of the popular vote in the state, in order to improve the efficiency of vote counting under the winner-notice method, the blockchain in the voting system consists of multiple candidate chains instead of a single chain of all ballot blocks linked together.

In the preparation phase of the election, the election administrator first generates the settings for the election, which contains basic information about the election such as the list of candidates, election id, election start time and end time. Then the election administrator broadcasts the settings to the blockchain network for voting, and the blockchain network receives them and generates the founding blocks for each candidate based on the information in the settings. As shown in Algorithm 2, the primary state election node collects the ballots from its state and verifies the compliance of the ballots.

Algorithm 2. Ballots collection and verification in *state*$_i$ election blockchain

Input: $ballots[]$, candidate list $Clist[]$, $verify()$: a function to verify the compliance of ballots
Output: candidate's vote set $BforC_i$

```
Begin:
for each ballot ∈ ballts[] do
    if verify(ballot) = True then
    for each C_i ∈ Clist[] do
        if ballot is voted for C_i then
            BforC_i.append(ballot);
        end if
    end for
    end if
end for
end
```

To prevent the elector node from being malicious, the ballot of the elector node contains the hash of the winning candidate chain. And since the winner-take-all method needs to be followed, all elector nodes should vote after the state election is over, and elector nodes in the same state should choose the same candidate, the state election blockchain will easily detect when an elector node is malicious or faulty.

V. VOTING PROCESS

The proposed voting scheme consists of several phases in an orderly manner as shown in Fig. 3.The voting process in the system. Each phase consists of one or more operations, and each operation is performed by one or more entities. The phases are described in detail below.

A. Assumptions

Our proposed electronic voting system operates on the following assumptions.

- Election administrator and national election authority are trusted and will not crash. And the above two institutions are operated under government supervision.
- Voters cast their ballots through the website or app. We assume that these apps are reliable and secure.

B. Preparation Phase

- First, the election administrator (EA) initializes the voting settings. EA first generates voting settings set_i (set_i=($Clist_i$, e_i,Start Timestamp, End Timestamp)) based on this election, including the election id (e_i), the list of candidates ($Clist_i$), and the start and end times of the election (Start Timestamp, End Timestamp).After that, EA signs the set_i with his own private key to generate the final settings $setting_i$ =(set_i , $Sign_{EA}$ (set_i)). Then EA sends the $setting_i$ to the blockchain network and NEA. The blockchains in the blockchain network generate the genesis block for each candidate as described in Section IV.
- Voter V_i logins in the voting system, then voter V_i generates public and private key pair (pk_i,sk_i) with local tool. After that, voter V_i sends (pk_i , e_i)and information about identity to NEA for authorization. If NEA verifies that voter V_i is eligible to vote in e_i, NEA will use its own private key to sign the (pk_i,e_i)as σ and return σ to the voter.

C. First Voting and tallying Phase

At this phase voters choose their favorite candidate and cast their ballots.

- The voter V_i first chooses the candidate C_i ($C_i \in Clist_i$)that he wants to vote for and then generates a ballot B_i=(C_i,pk_i,e_i,$state_i$,σ), then voter V_i uses the private key to sign the ballot B_i to get the final ballot $Ballot_i$ =(B_i , $Sign_{V_i}$ (B_i)), and finally casts $Ballot_i$ to the state election blockchain of corresponding state.
- The state election blockchain is responsible for packing the ballots into blocks, forming multiple candidate chains in the manner of Section IV, with each candidate chain linked only to the blockchain that holds the selection of that candidate.

D. Tallying and Second Voting Phase

- Since the ballots have formed multiple candidate chains in the state blockchain, the vote tallying is simple. After voters finish voting (i.e., the current timestamp is greater than the end timestamp), the elector nodes need to cast their votes according to the winner-take-all method, and then the elector nodes only need to get the height of each candidate chain through the corresponding state blockchain API. Elector nodes compare block chain heights to get election results, and then the candidate nodes will only vote for the candidate who corresponding to the

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highest height candidate chain. The process is as shown in Algorithm 4.

Algorithm 4 Tallying and second Voting Phase

Input: elector node in $state_i$, the ballot size of a block body k, $CandidateChains$ in $state_i$

Output: voting result in $state_i$

1: **Begin:**

2: **for** each $CandidateChains_i \in CandidateChains$ **do**

3: get $ChainHight_i$ of $CandidateChains_i$

4 $res_{C_i} = (ChainHight_i - 1) * k +$ the number of ballots in last block

5: **end for**

6: $C = Max(resC_i)$

7: choose winner candidate C for vote

8: $HashOfC = hash(\sum_{n=1}^{ChainHighC} HashOfBlock_n)$

9: elector node generates $B = (C_i, pk_i, e_i, state_i, HashOfC)$

10: send $ballot = (B, Sign_{ele}(B))$ to national election blockchain

11: **End**

E. Tallying and Publish the Result

Similarly, the national election blockchain only needs to compare the heights of candidate blockchains to get the results. The national blockchain is also responsible for publishing the election results, and when an organization or individual needs to audit the election results the national blockchain can publish the candidate chain.

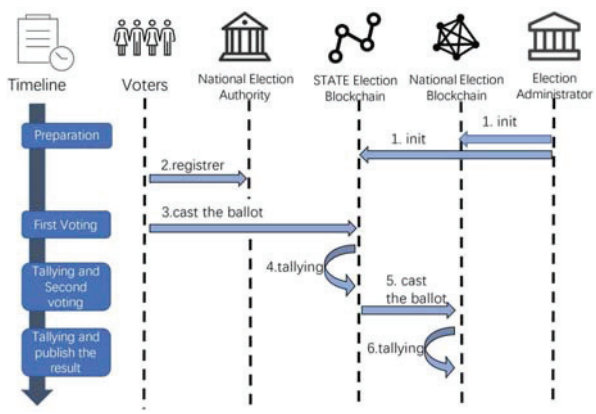

Fig. 2. The voting process in the system.

VI. Evaluation

A. Analysis of Evaluation Criteria

After studying the existing electronic voting models [22], we introduce the following common evaluation criteria that a blockchain-based secure electronic voting model must fulfill:

- Eligibility: The voter's identity needs to be verified through NEA before the voter can vote. This ensures that the voter is eligible. At the same time, the features of digital signature are utilized in the verification process to prevent forgery and tampering. Therefore, our voting system is able to meet the eligibility requirements.
- Correctness: This requirement simply means that the ballots should be counted correctly.
- Unreusability: Unreusability requires that no one be able to cast a vote twice. Our voting system will verify the compliance of each ballot and record the public key in the ballot. The blockchain will directly detect duplicate ballots and reject the ballot if the verification fails.
- Vote-and-Go: Our voting system does not need the voter to trigger the tallying phase. Moreover, in our system, voters can also cast their vote and quit before the voting ends.
- Affordability: Different from other voting systems based on cryptocurrency, our proposed voting system is built on the basis of consortium blockchain, and we redesigned the ballots so that cryptocurrency is not needed in the voting process, thus reducing the cost of election.

B. Performance analysis of the BCvoteMED

To evaluate the performance, the BCvoteMDE scheme was implemented on Hyperledger Fabric (version 2.2), the development environment was in Ubuntu 18.04. The number of candidates is 2.

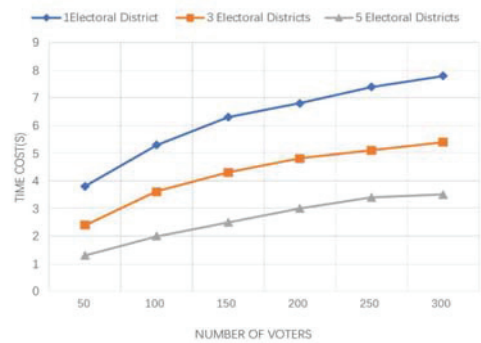

Fig. 3. The time cost with different number of electoral districts.

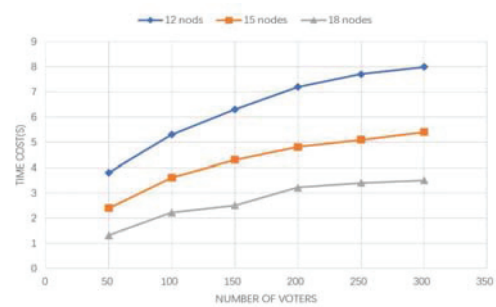

Fig. 4. The time cost with different number of state election node.

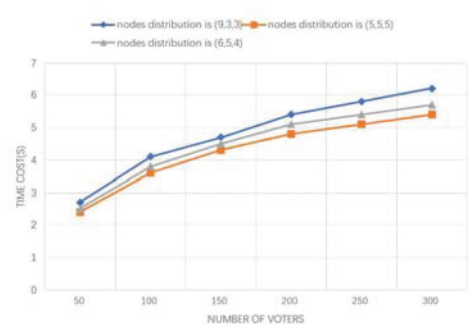

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Fig. 5. The time cost with different nodes distribution.

First of all, we test the time cost of with different number of electoral districts of BCvoteMDE in Fig3. The time cost is defined as the period between the casting of ballots and the announcement of voting results in our scheme. Groups of different number of voters cast their ballots according to electoral district, we set the number of electoral districts varying from 1 to 3. 15 nodes are equally distributed to each electoral districts. The number of upper blockchain nodes is fixed at 4. As shown in the Fig.3, the time cost increases with the increase of the number of voters, and the time cost increases with the increase of the number of electoral district under the same number of voters. Moreover, the introduction of electoral district shares the performance pressure.

Next, we test time cost with different number of state election node in Fig4. The number of total state election node varying from 12 to 15. We set the number of electoral districts to 3, and all nodes are equally distributed to each electoral districts. As shown in the Fig.4, it can be concluded that as the number of nodes in each district increases, the time also raises.

Moreover, we test time cost with different nodes distribution in Fig5. We set the number of electoral districts to 3, but the number of each districts is different. Fig.5 depicts the relationship between nodes distributing and time cost. The more evenly distributed the number of nodes, the less time consumption. This may be the difference of consensus time between different nodes.

VII. Conclusion

In this paper, we propose a new blockchain-based e-voting scheme for multi-district elections, and we work to apply an efficient and reliable blockchain-based e-voting system to election scenarios using the electoral college. In order to better implement the multi-district elections for elections, we design a two-layer blockchain architecture and describe a new voting process based on the multi-district based elections. The two-layer blockchain architecture meets the needs of the multi-district elections and improves the efficiency of voting.

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